Hasselt University

Project Learning from Data
3222–2526

Determinants of Household Electricity Consumption

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August 12, 2026

Abstract

This study investigates the determinants of annual residential electricity consumption using a random sample of 2,500 Belgian households. After group-specific row removal, data-entry error correction, and complete-case analysis, the final analytic sample comprised 2,477 households. Bidirectional stepwise AIC identified living area, EPC label, occupancy count, household income, and distance to Brussels as the best predictors (Adj. $R^2 = 0.853$, $F(9, 2467) = 1595$, $p < 0.001$). Each additional resident was associated with an increase of approximately 874 kWh per year. EPC label had a strong monotonic effect: label F households consumed 2,383 kWh more annually than label A households (95% CI: [2,253, 2,512], $p < 0.001$). Household income had no statistically significant independent effect once structural and occupancy predictors were controlled for ($p = 0.115$, $\beta^* = 0.018$). A significant interaction between EPC group and occupancy was detected ($F(1, 2470) = 86.03$, $p < 0.001$): the per-resident energy penalty is approximately 182 kWh larger in low- versus high-EPC homes, and this gap widens with household size.

1 Introduction

Residential buildings account for a substantial share of total EU energy consumption, making the housing sector central to achieving climate neutrality by 2050. In Belgium, the Energy Performance Certificate (EPC) system rates properties from A (most efficient) to F (least efficient). While structural factors such as living area clearly drive consumption, the role of income is theoretically ambiguous: wealthier households may invest in high-efficiency renovations yet simultaneously occupy larger homes. Disentangling these competing drivers is essential for evidence-based policy.

This study addresses four pre-specified research questions:

- **RQ1:** Which set of predictors best explains annual residential electricity consumption?
- **RQ2:** How does annual electricity use differ across EPC ratings B–F relative to label A?
- **RQ3:** To what extent does household income affect electricity consumption independently of structural and occupancy predictors?
- **RQ4:** Does the impact of a high-quality EPC rating (label A or B) change as household occupancy increases?

2 Data and Preprocessing

Three observations were removed per group-specific rules ($g = 7$): rows 49 (7^2), 77 (7×11), and 154 ($19 \times 7 + 21$). A range check on `annual_kwh` identified five observations with physically impossible values (2.9–5.4 million kWh), consistent with missing decimal separators during data entry; the highest legitimate value was 16,395 kWh. These were removed prior to any model fitting. Six variables each contained five missing values (0.2%), well below the pre-specified 10% threshold; complete-case analysis yielded a final analytic sample of $n = 2{,}477$.

The outcome was analysed on its original scale; no transformation was applied as OLS assumptions were sufficiently satisfied after data validation. Continuous predictors were retained in their original units and `occupancy_count` was treated as continuous, standard practice for integer counts with sufficient range (1–7). Categorical predictors were dummy-coded with reference categories: `epc_label` = A, `building_era` = Post-2000, and `roof_color` = Black. For RQ4, a binary variable `high_epc` was derived, coding labels A–B as High ($n = 989$) and labels C–F as Low ($n = 1{,}488$).

Table 1: Descriptive statistics for continuous variables ($n = 2{,}477$).

Variable	Mean	SD	Min	Max
`annual_kwh`	5,870	2,214	1,825	16,395
`sq_meters`	158.2	36.4	60.3	341.3
`income_euro`	48,816	22,500	8,488	206,368
`occupancy_count`	3.99	1.98	1.00	7.00
`dist_to_brussels`	62.9	33.0	5.1	120.0

The EPC label A was the most common (22.3%) and label F the least common (9.6%); the three construction eras were approximately equally represented (≈33% each).

3 Statistical Methods

3.1 Model specification and outcome scale

Let $Y_i = \texttt{annual\_kwh}_i$ denote the annual electricity consumption of household $i = 1, \dots, n$. The general multiple linear regression model is:

$$Y_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ij} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \tag{1}$$

where X_{ij} are the p predictors and ε_i are independently and identically distributed error terms. Following data validation, the full model was fitted on the clean dataset of n = 2,477 observations. Diagnostic plots indicated that OLS assumptions were sufficiently satisfied without transformation; coefficients are therefore interpreted directly in kWh units.

3.2 Model selection (RQ1)

Starting from a full model containing all eight predictors, bidirectional stepwise AIC was applied. At each step, the predictor whose removal or addition most reduced AIC was dropped or added until no further improvement was possible. Stepwise selection may produce unstable models and does not guarantee a globally optimal solution; it is used here for exploratory purposes consistent with the pre-specified analytical plan. As AIC prioritises predictive performance over inferential significance, predictors retained by the procedure may not necessarily reach $\alpha = 0.05$.

The incremental contribution of predictor blocks was assessed by comparing adjusted R^2 and AIC across nested model blocks: structural → socioeconomic → geographic → screening. For RQ2, $H_0\colon \beta_k = 0$ versus $H_1\colon \beta_k \neq 0$ was tested for each label $k \in \{B, C, D, E, F\}$ relative to label A; for RQ3, $H_0\colon \beta_{\texttt{income}} = 0$ versus $H_1\colon \beta_{\texttt{income}} \neq 0$.

No correction for multiple comparisons was applied to the five RQ2 contrasts, as these were pre-specified on a monotonic ordinal scale; all five remained significant under Bonferroni correction ($\alpha_{\text{adj}} = 0.01$). All tests were pre-specified at $\alpha = 0.05$ (two-sided).

3.3 Interaction model (RQ4)

To address RQ4, a binary variable high_epc was derived:

$$\texttt{high\_epc}_i = \begin{cases} 1 & \text{if } \texttt{epc\_label}_i \in \{A, B\} \\ 0 & \text{otherwise (labels C–F)} \end{cases}$$

The interaction model was then specified as:

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1\, \texttt{sq\_meters}_i + \beta_2\, \texttt{high\_epc}_i + \beta_3\, \texttt{occupancy\_count}_i \\ &+ \beta_4\, (\texttt{high\_epc}_i \times \texttt{occupancy\_count}_i) + \beta_5\, \texttt{income\_euro}_i + \beta_6\, \texttt{dist\_to\_brussels}_i + \varepsilon_i \end{aligned} \tag{2}$$

The interaction model was compared to the additive model via an F-test with one degree of freedom ($H_0\colon \beta_4 = 0$, $H_1\colon \beta_4 \neq 0$).

3.4 Model assumptions and diagnostics

The OLS assumptions were assessed as follows:

- **Linearity and homoscedasticity:** residuals versus fitted values and scale–location plots.
- **Normality:** Q–Q plot of standardised residuals against theoretical normal quantiles.
- **Multicollinearity:** Generalised Variance Inflation Factor (GVIF); threshold $\text{GVIF}^{1/(2\cdot\text{df})} < \sqrt{5} \approx 2.24$.
- **Outliers and influence:** leverage values via the hat matrix diagonal h_{ii}, threshold $h_{ii} > 2(p+1)/n$.

All analyses were performed in R 4.5.2 at $\alpha = 0.05$ (two-sided).

4 Results

4.1 RQ1: Determinants of Electricity Consumption

Starting from a full model containing all eight predictors, bidirectional stepwise AIC selection identified the following preferred model:

$$\begin{aligned} \texttt{annual\_kwh}_i = \beta_0 + \beta_1\,\texttt{sq\_meters}_i + \beta_2\,\texttt{epc\_label}_i + \beta_3\,\texttt{occupancy\_count}_i \\ + \beta_4\,\texttt{income\_euro}_i + \beta_5\,\texttt{dist\_to\_brussels}_i + \varepsilon_i \end{aligned} \tag{3}$$

Three predictors were eliminated sequentially: `roof_color` (step 1), `building_era` (step 2), and `pet_ownership` (step 3), leaving `income_euro` and `dist_to_brussels` retained by AIC despite being non-significant at $\alpha = 0.05$. Table 2 presents the model comparison across incremental blocks, and Table 3 presents the full coefficient table for the preferred model.

Table 2: Model comparison across incremental predictor blocks.

Model	Predictors	Adj. R^2	AIC
1 — Structural	`sq_meters`, `epc_label`, `building_era`	0.244	40,504.8
2 — Socioeconomic	+ `income_euro`, `occupancy_count`	0.853	40,457.1
3 — Geographic	+ `dist_to_brussels`	0.853	40,456.5
4 — Screening	+ `roof_color`, `pet_ownership`	0.853	40,460.8
Preferred	`sq_meters`, `epc_label`, `occupancy_count`, `income_euro`, `dist_to_brussels`	0.853	40,453.5

Table 3: Preferred model coefficient estimates (RSE = 849.3 kWh, Adj. $R^2 = 0.853$, $F(9,\ 2467) = 1595$, $p < 0.001$, $n = 2{,}477$).

Term	Estimate	Std. Error	t	p-value	95% CI
Intercept	−2,255	98.7	−22.85	< 0.001	[−2,448, −2,061]
`sq_meters`	22.9	0.68	33.63	< 0.001	[21.5, 24.2]
`epc_label` B	489.1	54.4	8.99	< 0.001	[382, 596]
`epc_label` C	883.7	51.8	17.06	< 0.001	[782, 985]
`epc_label` D	1,410	56.3	25.06	< 0.001	[1,300, 1,520]
`epc_label` E	1,945	58.8	33.09	< 0.001	[1,830, 2,060]
`epc_label` F	2,383	66.1	36.07	< 0.001	[2,253, 2,512]
`occupancy_count`	873.8	8.65	100.98	< 0.001	[857, 891]
`income_euro`	0.002	0.001	1.58	0.115	[−0.000, 0.004]
`dist_to_brussels`	−0.83	0.52	−1.61	0.107	[−1.85, 0.18]

Occupancy count is the dominant predictor ($t = 100.98$): each additional resident is associated with an increase of approximately 874 kWh per year. Living area contributes independently, with each additional m^2 adding 23 kWh. EPC label has a strong monotonic effect, discussed fully under RQ2. Income and distance to Brussels were retained by AIC but are not significant at $\alpha = 0.05$ (both $p > 0.10$, CIs crossing zero), suggesting their independent contribution is negligible once structural and occupancy predictors are accounted for.

4.2 Model Diagnostics

Residual diagnostic plots for the preferred model showed some mild curvature and variation in residual spread, but no severe departure from the assumptions required for the analysis. The Q–Q plot indicated approximately normal residuals with some deviation in the upper tail. Given the large sample size ($n = 2{,}477$), these deviations were considered acceptable for the present analysis. All $\text{GVIF}^{1/(2\cdot\text{df})}$ values were below 1.46, well within the threshold of $\sqrt{5} \approx 2.24$, indicating no substantial multicollinearity concerns. Leverage analysis identified 30 high-leverage observations ($h_{ii} > 2(p+1)/n = 0.0081$), representing 1.2% of the sample; no observations appeared to exert undue influence on model estimates (see Appendix E).

4.3 RQ2: Effect of EPC Ratings

The EPC coefficients reported in Table 3 show the adjusted differences in annual electricity consumption for labels B–F relative to the reference category (label A), holding living area, occupancy, income, and distance to Brussels constant. All five labels differed significantly from label A ($p < 0.001$). The effect increases monotonically with declining energy efficiency: label B households consume approximately 489 kWh more per year than label A households, rising to 2,383 kWh more for label F. The sharpest single step occurs between labels D and E (+535 kWh).

4.4 RQ3: Income Effect

Although `income_euro` was retained by stepwise AIC, it did not reach statistical significance ($\hat{\beta} = 0.0017$, SE = 0.0011, $t = 1.58$, $p = 0.115$, 95% CI: $[-0.000,\ 0.004]$). Its standardised coefficient ($\beta^* = 0.018$) was negligible compared to occupancy count ($\beta^* = 0.781$) and living area ($\beta^* = 0.376$). The bivariate correlation between income and consumption ($r = 0.288$) is largely explained by the strong correlation between income and living area ($r = 0.724$): wealthier households occupy larger homes, and it is home size rather than income that drives higher consumption.

4.5 RQ4: Interaction Between EPC Group and Occupancy

To assess whether the energy-saving advantage of a high EPC rating changes with household occupancy, an interaction model was fitted allowing the per-resident electricity slope to differ between high-EPC homes (labels A–B, $n = 989$) and low-EPC homes (labels C–F, $n = 1{,}488$). An F-test comparing the interaction model against the additive model provided strong evidence for the interaction ($F(1, 2470) = 86.03$, $p < 0.001$; see Appendix D), indicating that the effect of EPC group on electricity consumption is not constant across occupancy levels.

Table 4 presents the interaction model coefficients. In low-EPC homes, each additional resident is associated with an increase of 941 kWh per year ($p < 0.001$). In high-EPC homes, this slope is significantly reduced by 182 kWh per additional resident ($\hat{\beta}_4 = -182.3$, 95% CI: $[-221.5, -143.1]$, $p < 0.001$), yielding a net increase of approximately 759 kWh per resident. The gap between high- and low-EPC homes therefore widens as occupancy increases, as illustrated in Figure 1.

Table 4: Interaction model coefficients for RQ4 ($n = 2{,}477$, RSE = 947 kWh, Adj. $R^2 = 0.817$).

Term	**Estimate**	**Std. Error**	t	p**-value**
Intercept	−981.7	107.8	−9.11	< 0.001
`sq_meters`	22.61	0.76	29.81	< 0.001
`high_epc` High	−553.0	87.9	−6.29	< 0.001
`occupancy_count`	941.2	12.4	75.74	< 0.001
`income_euro`	0.0017	0.001	1.41	0.159
`dist_to_brussels`	−0.76	0.58	−1.32	0.186
`high_epc` × `occupancy_count`	−182.3	19.7	−9.28	< 0.001

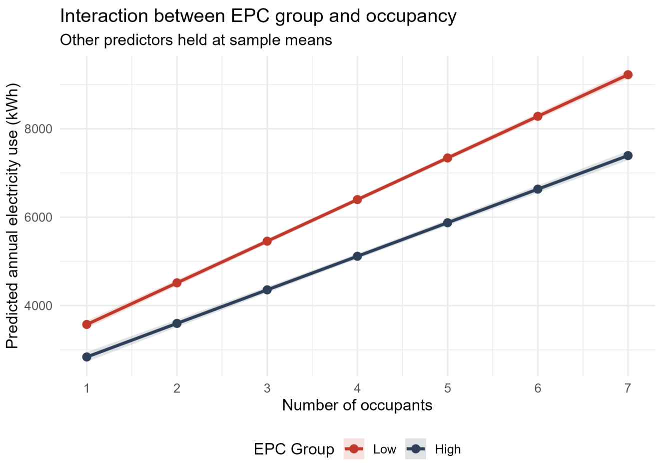

Figure 1: Predicted annual electricity consumption by EPC group and occupancy level. Diverging lines illustrate the significant interaction ($F(1,\, 2470) = 86.03$, $p < 0.001$); shaded bands are 95% CIs. Other predictors are held at sample means.

5 Discussion

Occupancy and living area were the dominant predictors of electricity consumption, confirming that structural and behavioural factors outweigh socioeconomic ones. The strong EPC gradient confirms the system captures real performance differences, with the largest step between labels D and E suggesting renovation of the lowest-rated properties offers the greatest return. Income disappears once home size is controlled, consistent with full mediation by living area rather than independent income-driven behaviour. The significant interaction indicates that high-EPC homes deliver increasing savings as household size grows, making efficiency improvements proportionally more valuable for larger households.

Limitations include the cross-sectional design, which prevents causal inference, and the retention of income and distance to Brussels by AIC despite non-significance at $\alpha = 0.05$.

6 Conclusion

This study examined the determinants of annual residential electricity consumption among 2,477 Belgian households using multiple linear regression. Bidirectional stepwise AIC identified living area, EPC label, occupancy count, income, and distance to Brussels as the best predictors (Adj. $R^2 = 0.853$, RQ1). EPC label had a strong monotonic effect, with label F households consuming 2,383 kWh more annually than label A households (RQ2). Household income had no significant independent effect on consumption ($p = 0.115$) once structural and occupancy predictors were controlled for (RQ3). A significant interaction between EPC group and occupancy was detected ($p < 0.001$): the energy-saving advantage of high-EPC homes increases with household size (RQ4). These findings suggest that policy efforts should prioritise EPC improvements and occupancy-aware interventions rather than income-based targeting.

7 Ethical Considerations

The findings carry several ethical implications for policy and stakeholders. First, since low-income households are most likely to occupy low-EPC homes yet least able to finance renovations, policies mandating energy efficiency upgrades risk deepening socioeconomic inequalities without targeted financial support. Second, consumption-based instruments such as tiered tariffs should account for household size, as penalising high consumption without controlling for occupancy would disproportionately burden larger families. Third, the household-level data used here — linking consumption to income, building characteristics, and location — raises privacy concerns; any operational deployment of such models requires informed consent and compliance with data protection legislation. Finally, the cross-sectional design precludes causal inference, and findings may not generalise beyond the Belgian regulatory context.

Portfolio Note

This report is a portfolio version of work developed in the context of the Project Learning from Data course at Hasselt University. The statistical analysis, R implementation, interpretation, visualisation, and report preparation presented in this version were carried out by the author.

Use of AI Tools

Claude (Anthropic) was used to assist with structuring the report, improving clarity of writing, and formatting LaTeX code. All statistical analyses, model implementation, and interpretation of results were conducted independently by the author using R 4.5.2.

References

[1] Thas, O. (2023). *Linear models.* Retrieved from https://othas.github.io/LIMO/ (accessed April 2026)

[2] Thas, O. (2025). *Project Learning from Data: Ethics.* Course lecture slides, UHasselt.

[3] Pieters, Z. (2025). *Writing a statistical report.* Course lecture slides, Project Learning from Data, UHasselt.

Appendix

A. Group-Specific Row Removal

Table 5: Group-specific row removal rules ($g = 7$).

Row	**Rule**	**Calculation**	**Removed**
49	g^2	$7^2 = 49$	Yes
77	$g \times 11$	$7 \times 11 = 77$	Yes
154	$19g + 21$	$19(7) + 21 = 154$	Yes

B. Categorical Variable Distributions

Table 6: Distribution of categorical variables ($n = 2{,}477$).

Variable	**Category**	N	**%**
EPC Label	A (best)	553	22.3
	B	436	17.6
	C	526	21.2
	D	388	15.7
	E	336	13.6
	F (worst)	238	9.6
Building Era	Post-2000	809	32.7
	1970–2000	850	34.3
	Pre-1970	818	33.0
Pet Ownership	No	1,496	60.4
	Yes	981	39.6
Roof Colour	Black	826	33.3
	Red	870	35.1
	Other	781	31.5

C. Standardised Coefficients

Table 7: Standardised coefficients for continuous predictors in the preferred model. Higher values indicate greater relative importance.

Predictor	**Standardised** β^*
`occupancy_count`	0.781
`sq_meters`	0.376
`income_euro`	0.018
`dist_to_brussels`	0.012

D. Interaction Model F-test

Table 8: F-test comparing additive and interaction models (RQ4).

Model	**Res. df**	**RSS**	**df**	F	p**-value**
Additive	2,471	2,292,477,662			
Interaction	2,470	2,215,321,851	1	86.03	< 0.001

E. Diagnostic Plots

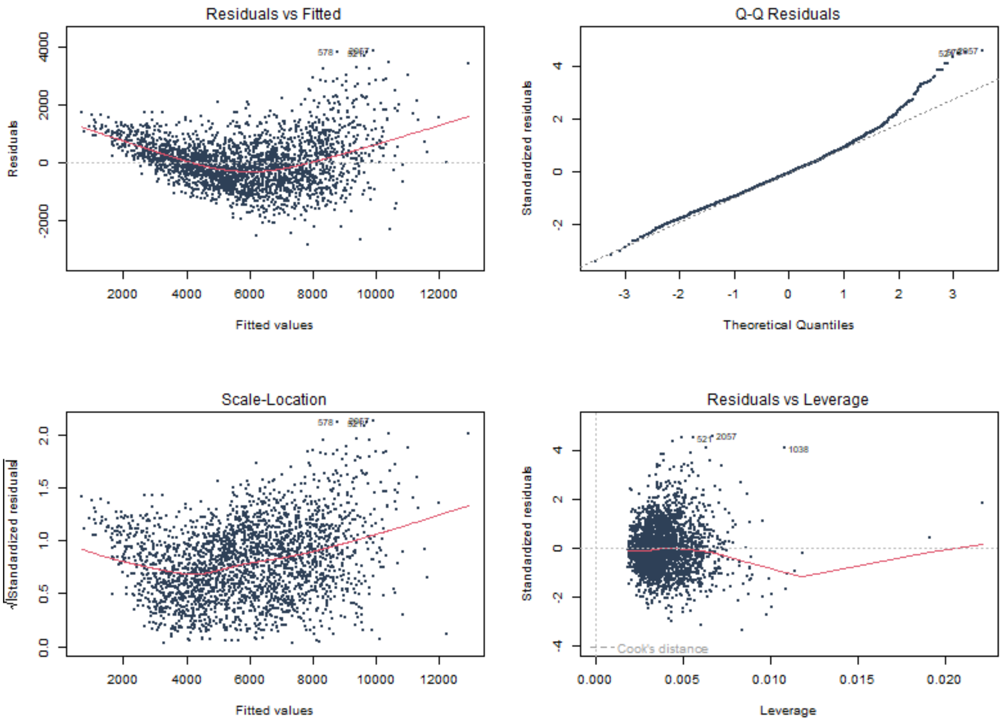

Figure 2: Diagnostic plots for the preferred model ($n = 2{,}477$). *Residuals vs Fitted*: mild curvature is present, with no severe departure from the fitted relationship. *Q–Q plot*: residuals follow the theoretical normal line reasonably well, with some upper-tail deviation. *Scale–Location*: some variation in residual spread is visible. *Residuals vs Leverage*: no observations appear to exert undue influence on the model estimates.